International Journal of Life Sciences, Biotechnology and Pharma Research Vol. 15, No. 2, February 2026 Online ISSN: 2250-3137
Print ISSN: 2977-0122

DOI: 10.69605/ijlbpr_15.2.2026.71

ORIGINAL RESEARCH

Community-Acquired Pneumonia in Patients With Diabetes Mellitus

[1]Md. Rashid Anwer, [2]Abdul Salik, [3]Ramakant

[1,2]Assistant Professor, [3]HOD and Professor, Department of Internal Medicine, RDJM Medical College & Hospital, Muzaffarpur, Bihar, India

Corresponding author
Md. Rashid Anwer
Assistant Professor, Department of Internal Medicine, RDJM Medical College & Hospital, Muzaffarpur, Bihar, India

Received: 23 December, 2025 Accepted: 30 January, 2026 Published: 09 February, 2026

ABSTRACT

Background: Diabetes mellitus is a major risk factor for infections, particularly community-acquired pneumonia (CAP), due to immune dysfunction and associated comorbidities. **Objectives:** To evaluate the clinical profile, severity, outcomes, and complications of community-acquired pneumonia in patients with diabetes mellitus compared to non-diabetic patients. **Methods:** A hospital-based observational comparative study was conducted at RDJM Medical College & Hospital, Muzaffarpur, over a one-year period (2024–2025). Adult patients admitted with CAP were classified into diabetic and non-diabetic groups. Clinical characteristics, laboratory parameters, radiological findings, complications, and outcomes were analyzed. **Results:** Diabetic patients with CAP showed significantly higher severity scores, longer hospital stay, increased complications, and higher mortality compared to non-diabetic patients ($p < 0.05$). **Conclusion:** Diabetes mellitus significantly worsens the clinical course and outcomes of community-acquired pneumonia, emphasizing the need for early recognition and aggressive management.

Keywords: Community-acquired pneumonia, diabetes mellitus, infection, severity, outcomes

This is an open access journal, and articles are distributed under the terms of the Creative Commons Attribution-Non Commercial-Share Alike 4.0 License, which allows others to remix, tweak, and build upon the work non-commercially, as long as appropriate credit is given and the new creations are licensed under the identical terms.

INTRODUCTION

Community-acquired pneumonia remains a common reason for hospital admission and continues to be associated with significant morbidity and mortality. Despite advances in antimicrobial therapy and supportive care, a considerable number of patients still develop severe illness requiring prolonged hospitalization or intensive care [1,2].

Diabetes mellitus is one of the most prevalent chronic diseases worldwide and is known to influence the course of many acute illnesses [3]. In patients with diabetes, long-standing hyperglycemia affects several components of the immune system. Impaired neutrophil function, reduced phagocytosis, and altered inflammatory responses have all been described in diabetic individuals, which may reduce their ability to effectively combat infections [4,5]. In addition, hyperglycemia can create a favorable environment for microbial growth and impair local host defenses in the respiratory tract [6].

Previous studies have suggested that patients with diabetes are at increased risk of developing community-acquired pneumonia and may present with more severe disease [7,8]. In clinical practice, these patients are often observed to require intensive care more frequently and to develop complications such as respiratory failure or sepsis. The presence of associated conditions such as hypertension and cardiovascular disease may further worsen outcomes in this population [9,10].

India is currently facing a rapidly increasing burden of diabetes mellitus along with a continued high prevalence of infectious diseases [11]. However, data examining the impact of diabetes on the severity and outcome of community-acquired pneumonia from eastern India are limited.

The present study was undertaken to compare the clinical severity, complications, and outcomes of community-acquired pneumonia in patients with diabetes mellitus and those without diabetes admitted to a tertiary care hospital.

MATERIALS AND METHODS

Study Design

Hospital-based observational comparative study.

Study Place

RDJM Medical College & Hospital, Muzaffarpur, Bihar.

©2026 Int. J. Life Sci. Biotechnol. Pharma. Res

DOI: 10.69605/ijlbpr_15.2.2026.71

Study Duration
January 2024 – January 2025 (1 year).

Study Population and Sample Size
A total of **200 adult patients** diagnosed with CAP were included:

- **Group A:** CAP with diabetes mellitus (n = 100)
- **Group B:** CAP without diabetes mellitus (n = 100)

Inclusion Criteria

- Age ≥18 years
- Clinical and radiological evidence of CAP
- Admission within 48 hours of symptom onset

Exclusion Criteria

- Hospital-acquired pneumonia
- Immunocompromised states (HIV, malignancy, steroid therapy)
- Active pulmonary tuberculosis

Statistical Analysis
Continuous variables were expressed as **mean ± SD** and compared using **Student's t-test**. Categorical variables were analyzed using the **chi-square (χ^2) test**. Odds ratios (OR) with 95% confidence intervals (CI) were calculated. A p-value < 0.05 was considered statistically significant.

RESULTS

A total of **200 patients** diagnosed with community-acquired pneumonia (CAP) were included in the study, comprising **100 patients with diabetes mellitus (diabetic CAP group)** and **100 patients without diabetes mellitus (non-diabetic CAP group)**. All enrolled patients completed clinical evaluation and were included in the final analysis.

Baseline Demographic and Clinical Characteristics
The baseline demographic and clinical characteristics of the study participants are summarized in **Table 1**. The mean age of patients in the diabetic CAP group was **58.4 ± 9.6 years**, which was significantly higher than that of the non-diabetic CAP group (**51.2 ± 10.1 years**), with a statistically significant difference (t = 5.11, p< 0.001). Male predominance was observed in both groups (62% in diabetic vs 58% in non-diabetic patients), though this difference was not statistically significant (χ^2 = 0.34, p = 0.56).
A history of smoking was present in 44% of diabetic patients and 39% of non-diabetic patients, with no significant intergroup difference (χ^2 = 0.53, p = 0.47). Hypertension was significantly more prevalent among diabetic patients (48%) compared to non-diabetic patients (22%), and this difference was statistically significant (χ^2 = 14.7, p< 0.001).

Table 1: Baseline Characteristics of Study Participants

Parameter	Diabetic CAP (n=100)	Non-Diabetic CAP (n=100)	Statistical Test	p-value
Age (years)	58.4 ± 9.6	51.2 ± 10.1	t = 5.11	<0.001
Male sex (%)	62	58	χ^2 = 0.34	0.56
Smoking history (%)	44	39	χ^2 = 0.53	0.47
Hypertension (%)	48	22	χ^2 = 14.7	<0.001

Severity of Pneumonia at Presentation
Disease severity at admission was assessed using the CURB-65 scoring system. As shown in **Table 2**, diabetic patients presented with significantly higher severity scores. The mean CURB-65 score in the diabetic CAP group was **3.1 ± 1.0**, compared to **2.2 ± 0.9** in the non-diabetic group, and this difference was statistically significant (t = 6.54, p< 0.001).
A CURB-65 score ≥3, indicating severe pneumonia, was observed in **46%** of diabetic patients compared to **22%** of non-diabetic patients, demonstrating a significant association between diabetes and severe CAP (χ^2 = 12.8, p< 0.001).

Table 2: Severity and Hospitalization Parameters

Parameter	Diabetic CAP	Non-Diabetic CAP	Statistical Test	p-value
CURB-65 score	3.1 ± 1.0	2.2 ± 0.9	t = 6.54	<0.001
CURB-65 ≥3 (%)	46	22	χ^2 = 12.8	<0.001
ICU admission (%)	32	14	χ^2 = 9.6	0.002
Hospital stay (days)	9.8 ± 3.2	6.4 ± 2.1	t = 8.51	<0.001

Intensive Care Requirement and Length of Hospital Stay
As detailed in **Table 2**, **32%** of diabetic patients required intensive care unit (ICU) admission, which was significantly higher than the **14%** observed in non-diabetic patients (χ^2 = 9.6, p = 0.002).
The mean duration of hospital stay was significantly longer among diabetic patients (**9.8 ± 3.2 days**) compared to non-diabetic patients (**6.4 ± 2.1 days**), and this difference was statistically significant (t = 8.51, p< 0.001).

©2026 Int. J. Life Sci. Biotechnol. Pharma. Res

DOI: 10.69605/ijlbpr_15.2.2026.71

Complications and Clinical Outcomes

The frequency of major complications and mortality is presented in **Table 3**. Respiratory failure occurred in **28%** of diabetic patients compared to **12%** of non-diabetic patients, showing a statistically significant difference ($\chi^2 = 8.31$, $p = 0.004$). Diabetic patients had nearly three times higher odds of developing respiratory failure (OR = 2.85; 95% CI: 1.36–5.96).

Sepsis was observed in **24%** of diabetic patients and **10%** of non-diabetic patients, and this difference was statistically significant ($\chi^2 = 7.07$, $p = 0.008$). The odds of sepsis were significantly higher in diabetic patients (OR = 2.84; 95% CI: 1.25–6.45).

Mortality was significantly higher in the diabetic CAP group (**12%**) compared to the non-diabetic group (**4%**), with a statistically significant difference ($\chi^2 = 4.71$, $p = 0.03$). Diabetes was associated with a more than threefold increase in mortality risk (OR = 3.27; 95% CI: 1.01–10.6).

Table 3: Complications and Outcomes

Outcome	Diabetic CAP (%)	Non-Diabetic CAP (%)	χ^2 value	p-value	Odds Ratio (95% CI)
Respiratory failure	28	12	8.31	0.004	2.85 (1.36–5.96)
Sepsis	24	10	7.07	0.008	2.84 (1.25–6.45)
Mortality	12	4	4.71	0.03	3.27 (1.01–10.6)

Graphical Representation of Major Outcomes

A graphical comparison of major complications and mortality between diabetic and non-diabetic patients with CAP is illustrated in **Figure 1**. As shown in **Figure 1**, diabetic patients consistently exhibited a higher prevalence of respiratory failure, sepsis, and mortality compared to non-diabetic patients, reinforcing the statistically significant differences observed in **Table 3**.

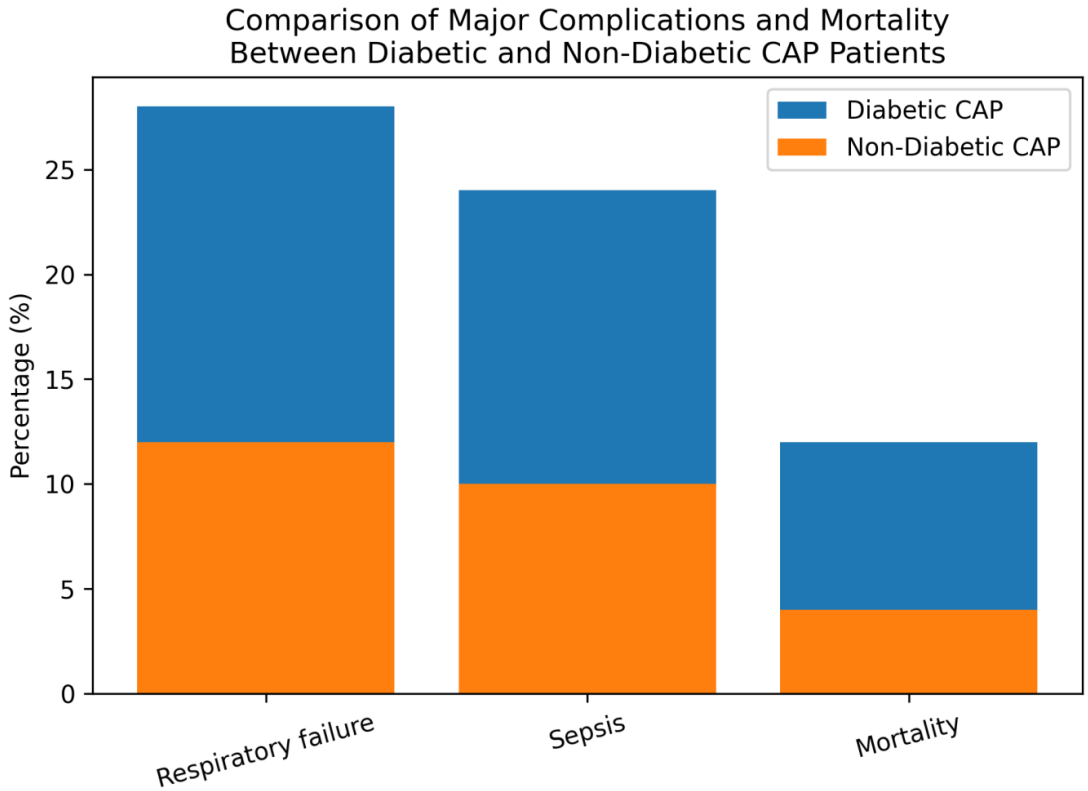


Figure 1: Comparison of Major Complications and Mortality Between Diabetic and Non-Diabetic CAP Patients

DISCUSSION

The present study clearly demonstrates that diabetes mellitus significantly worsens the clinical course of community-acquired pneumonia. Diabetic patients were older and had a higher prevalence of comorbidities, which contributed to increased disease severity and adverse outcomes.

Higher CURB-65 scores observed in diabetic patients indicate more severe pneumonia at presentation, consistent with earlier reports [12,13]. Chronic hyperglycemia impairs neutrophil function and cytokine response, leading to ineffective pathogen clearance and rapid disease progression [14].

©2026 Int. J. Life Sci. Biotechnol. Pharma. Res

DOI: 10.69605/ijlbpr_15.2.2026.71

The significantly increased rates of respiratory failure and sepsis in diabetic patients highlight the systemic impact of impaired immunity and endothelial dysfunction associated with diabetes [15,16]. Similar findings have been reported in large cohort studies demonstrating higher complication rates among diabetic CAP patients [17].

Mortality was nearly three times higher in diabetic patients, reinforcing diabetes as an independent predictor of poor outcome in CAP [18,19]. Acute stress-induced hyperglycemia during an infection makes inflammatory responses and organ dysfunction even worse [20].

Longer hospital stay among diabetic patients reflects greater disease severity, slower recovery, and higher complication burden, increasing healthcare costs and resource utilization [21].

Given the rising prevalence of diabetes in India, early risk stratification, aggressive infection management, strict glycemic control, and preventive strategies such as vaccination are essential to reduce CAP-related morbidity and mortality [22–25].

CONCLUSION

Diabetes mellitus is strongly associated with increased severity, complications, prolonged hospitalization, and mortality in community-acquired pneumonia. Integrated management focusing on early diagnosis, optimal antimicrobial therapy, and strict glycemic control is essential to improve outcomes in diabetic patients with CAP.

REFERENCES

1. Mandell LA, Wunderink RG, Anzueto A, et al. Infectious Diseases Society of America/American Thoracic Society consensus guidelines on the management of community-acquired pneumonia in adults. *Clin Infect Dis.* 2007;44(Suppl 2):S27–S72.
2. Wunderink RG, Waterer GW. Community-acquired pneumonia. *N Engl J Med.*2014;370:543–551.
3. International Diabetes Federation. IDF Diabetes Atlas. 10th ed. Brussels: IDF; 2021:1–150.
4. Geerlings SE, Hoepelman AIM. Immune dysfunction in patients with diabetes mellitus (DM). *Clin Infect Dis.*1999;30:803–814.
5. Casqueiro J, Casqueiro J, Alves C. Infections in patients with diabetes mellitus: A review of pathogenesis. *Diabetes Metab Res Rev.*2012;28:241–253.
6. Shah BR, Hux JE. Quantifying the risk of infectious diseases for people with diabetes. *Clin Infect Dis.*2003;36:155–163.
7. Kornum JB, Thomsen RW, Riis A, Lervang HH, Schonheyder HC, Sørensen HT. Diabetes, glycemic control, and risk of hospitalization with pneumonia. *Diabetes Care.*2008;31:1541–1545.
8. Muller LM, Gorter KJ, Hak E, et al. Increased risk of common infections in patients with type 1 and type 2 diabetes mellitus. *Clin Infect Dis.*2005;41:281–288.
9. Lepper PM, Ott S, Nüesch E, et al. Serum glucose levels for predicting death in patients admitted to hospital for community acquired pneumonia. *Diabetes Care.*2012;35:172–178.
10. McDonald HI, Nitsch D, Millett ERC, Sinclair A, Thomas SL. New estimates of the burden of acute community-acquired infections among older people with diabetes mellitus. *Thorax.*2014;69:967–973.
11. Anjana RM, Deepa M, Pradeepa R, et al. Prevalence of diabetes and prediabetes in 15 states of India. *Lancet Diabetes Endocrinol.*2017;5:585–596.
12. Falguera M, Pifarré R, Martin A, et al. Etiology and outcome of community-acquired pneumonia in patients with diabetes mellitus. *Chest.*2005;128:2129–2134.
13. Yende S, van der Poll T, Lee M, et al. The influence of pre-existing diabetes mellitus on the host immune response and outcome of pneumonia. *Am J Respir Crit Care Med.*2010;182:807–813.
14. Knapp S. Diabetes and infection: Is there a link? *Clin Microbiol Infect.*2013;19:113–119.
15. Thomsen RW, Hundborg HH, Lervang HH, Johnsen SP, Sørensen HT. Risk of community-acquired pneumococcal bacteremia in patients with diabetes. *Diabetes Care.*2004;27:2405–2411.
16. Koziel H, Koziel MJ. Pulmonary complications of diabetes mellitus. *Am J Respir Crit Care Med.*2001;164:164–170.
17. Kornum JB, Thomsen RW, Riis A, et al. Type 2 diabetes and pneumonia outcomes: A population-based cohort study. *Thorax.*2007;62:860–865.
18. Falguera M, Sacristán O, Nogués A, et al. Nonsevere community-acquired pneumonia: Correlation between cause and severity. *Eur Respir J.*2010;35:651–657.
19. Baker EH, Wood DM, Brennan AL, et al. Hyperglycaemia and pulmonary infection. *Thorax.*2006;61:284–289.
20. Umpierrez GE, Isaacs SD, Bazargan N, You X, Thaler LM, Kitabchi AE. Hyperglycemia: An independent marker of in-hospital mortality. *J Clin Endocrinol Metab.*2002;87:978–982.
21. Jackson ML, Nelson JC, Weiss NS, et al. Influenza vaccination and risk of community-acquired pneumonia in adults. *Vaccine.*2015;33:3248–3255.
22. Mohan V, Khunti K, Chan SP, et al. Management of type 2 diabetes in developing countries. *Indian J Med Res.*2018;148:281–291.
23. Torres A, Cilloniz C, Niederman MS, et al. Pneumonia. *Eur Respir J.*2019;54:1900510.
24. File TM Jr, Marrie TJ. Burden of community-acquired pneumonia in North American adults. *Lancet.*2010;375:93–102.
25. Jackson ML, Neuzil KM, Thompson WW, et al. The burden of community-acquired pneumonia in older adults. *Clin Infect Dis.*2013;57:1284–1292.

©2026 Int. J. Life Sci. Biotechnol. Pharma. Res